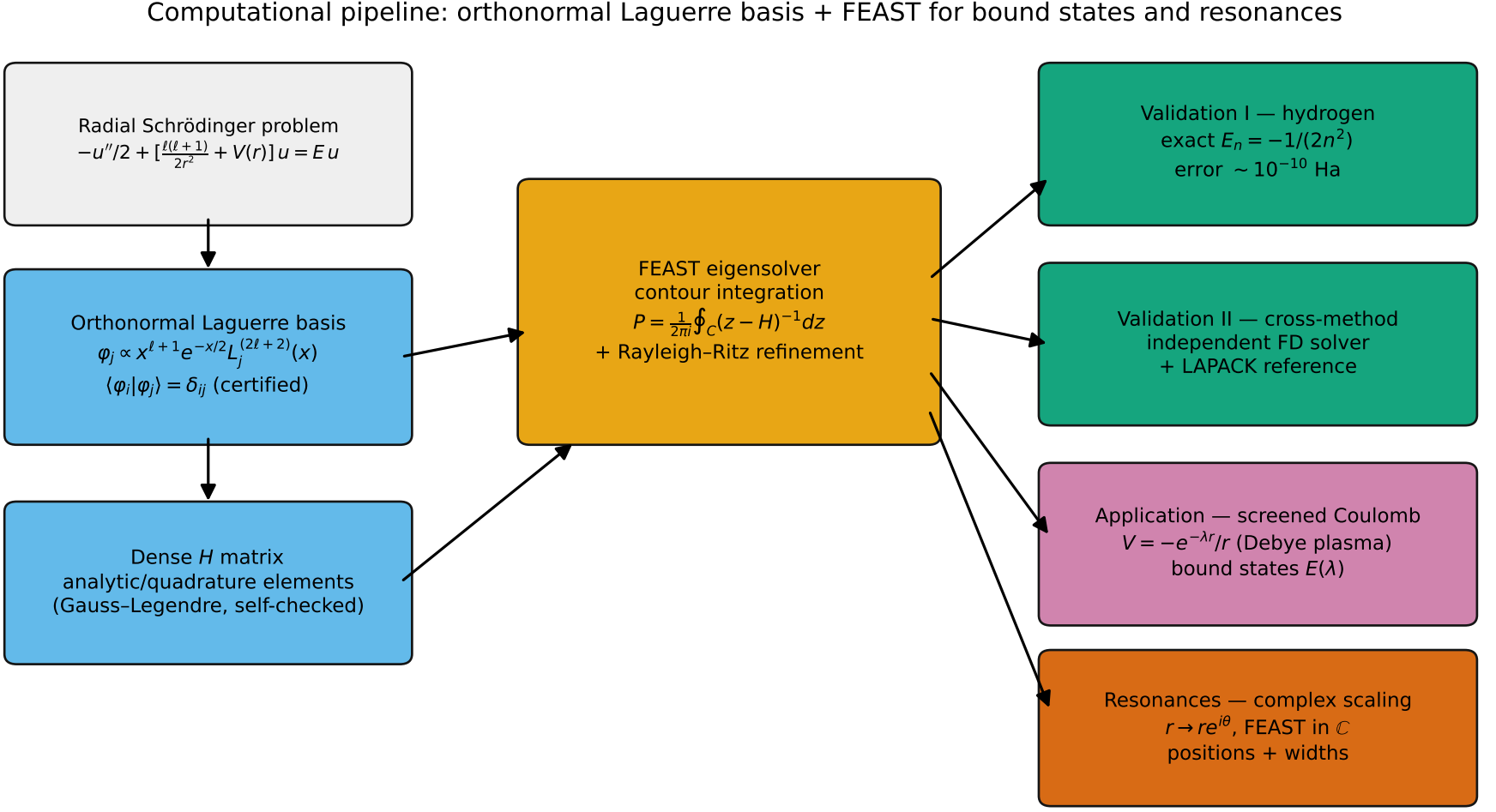
Computational pipeline: orthonormal Laguerre basis + FEAST for bound states and resonances
Radial Schrödinger problem
$-u''/2 + [\frac{\ell(\ell+1)}{2r^2} + V(r)]\,u = E\,u$
Orthonormal Laguerre basis
$\varphi_j \propto x^{\ell+1} e^{-x/2} L_j^{(2\ell+2)}(x)$
$\langle\varphi_i|\varphi_j\rangle = \delta_{ij}$ (certified)
Dense H matrix
analytic/quadrature elements
(Gauss–Legendre, self-checked)
FEAST eigensolver
contour integration
$P = \frac{1}{2\pi i}\oint_C (z-H)^{-1} dz$
+ Rayleigh–Ritz refinement
Validation I — hydrogen
exact $E_n = -1/(2n^2)$
error $\sim 10^{-10}$ Ha
Validation II — cross-method
independent FD solver
+ LAPACK reference
Application — screened Coulomb
$V = -e^{-\lambda r}/r$ (Debye plasma)
bound states $E(\lambda)$
Resonances — complex scaling
$r \to re^{i\theta}$, FEAST in $\mathbb{C}$
positions + widths